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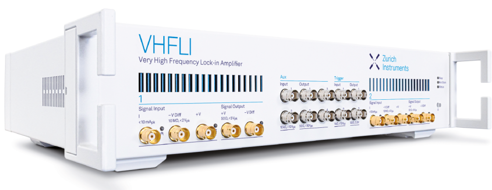

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ABSTRACT

In this work, the dynamics and formation mechanisms of Z-pinch plasma in a coaxial plasma gun were investigated. By using arrays of magnetic probes, calorimeters, and ballistic pendulums, the profiles of current and Lorentz force density, plasma momentum, and kinetic energy were detected. It was revealed that the flow Z-pinch was governed by a radial compression process consisting of two distinct stages. After being ejected from the inter-electrode gap, the plasma was first concentrated in the near-axis region downstream of the inner electrode and subsequently compressed by the pinching force generated by the axial current, resulting in the formation of a flow Z-pinch. A more pronounced concentration of momentum and energy, favorable for pulsed plasma spraying, was achieved by increasing the capacitor charging voltage and decreasing the gas injection time. A lower but spatially uniform absorbed energy density, suitable for film deposition and plasma etching, was obtained under a prolonged gas injection time. These results elucidated the Z-pinch plasma formation mechanisms in coaxial plasma guns and provided a reference for modulating plasma parameters in material processing applications.

I. INTRODUCTION

Plasma jets produced by coaxial plasma guns had numerous promising applications, including fusion energy,[1–4] plasma propulsion,[5,6] surface modification,[7,8] neutron generation,[9] and laboratory astrophysics.[10] As a typical feature of discharge in coaxial plasma guns, Z-pinch was formed by the radial compression of the current path, which caused an increase in the electron density and the energy density by more than one order of magnitude and thereby extended the range of its applications. For instance, the plasma jet of high number density ($\sim 10^{23}\,\mathrm{m^{-3}}$) enabled the study of collisional effects during plasma–surface interactions.[11] A high-energy density ($\sim 0.1\,\mathrm{MJ/m^2}$) pinched plasma region allowed processes such as quenching of steels and fabrication of submicrometer structures on silicon surfaces.[12] Moreover, the kink instability of the pinched plasma had been identified as a mechanism for magnetic flux amplification, which was crucial for spheromak formation.[13] As a result, the Z-pinch effect in coaxial plasma guns has received considerable attention over the past decades. Shumlak *et al.*[14] found that the Z-pinch could be stabilized by sufficient sheared axial flow, maintaining the plasma in high electron density ($\sim 10^{23}\,\mathrm{m^{-3}}$) and high electron temperature ($\sim 1\,\mathrm{keV}$) for tens of microseconds, in order to be candidates for controlled fusion reactors. Underwood *et al.*[15] identified the dynamic characteristics of coherent flow in the pinched plasma jet through the Schlieren method, and the flow velocities were measured by tracking these characteristics. Loebner *et al.*[16] computed the radial plasma density profile in the pinched region from Vogit fits of the H_α emission profile and predicted the temperature by the constant drift velocity (CDV) equilibrium Z-pinch model. The consistency of the temperature calculated from Doppler broadening of impurity emission and the predicted one provided strong evidence for the formation of a radially equilibrated Z-pinch. Giovannini *et al.*[17] measured the ion current density profile of the plasma jet from a magneto-plasma compressor (MPC) by using an array of Faraday cups and reported that the application of an external magnetic field increased the mean ion charges and electron temperature, thereby enhancing the photon emission.

Although it was established that the Z-pinch arose from the radial Lorentz force generated by the interaction of axial current with the self-induced azimuthal magnetic field, the underlying formation mechanisms remained systematically unexplored. However, not all coaxial plasma gun configurations were suitable for Z-pinch production. Working gas distribution within the gun was a critical factor as well. For example, in the FuZE device, a short gas injection time with low injected mass led to a discharge of deflagration mode and formed a collimated pinch.[18] In the SPG facility, the reduction of brightness of the pinched plasma was observed from ICCD images with a longer injection time.[19] Also, an excessively long inner electrode could even suppress the formation of a Z-pinch.[20] Until now, the relationships between the pinch effect and the operating parameters of discharge and geometry of coaxial plasma gun are still unclear; more work needs to be done to clarify the process and mechanisms of pinch formation.

Moreover, the compression of the plasma flow enabled the generation of high heat flux density ($\sim$50 GW/m^2) using capacitor banks with low stored energy (<10 kJ), rendering coaxial plasma guns efficient tools for material processing. However, energy density and momentum density of the pinched plasma, which are critical parameters for impulse plasma deposition (IPD)[21] and pulsed plasma spraying (PPS),[22] have seldom been reported.

In this work, the mechanisms of Z-pinch plasma formation in a gas-puff coaxial plasma gun were investigated. Profiles of the Lorentz force near the muzzle, where the pinch initiates, were computed from the azimuthal magnetic field measured by an array of magnetic probes. The observation of centrally peaked energy density and momentum density profiles provided strong evidence of the radial compression of the flowing plasma. The influence of operating conditions on pinch formation was evaluated. In the end, potential applications of the plasma jet in materials processing and thermal spray technology were discussed.

The organization of this paper is as follows. Section II describes the setup of the plasma gun and the diagnostics. Section III presents the experimental results and discusses the physical mechanisms and potential applications. Section IV gives the conclusion and overview of the results.

II. EXPERIMENTAL SETUP AND DIAGNOSTICS

A. Experimental setup

A schematic of the experimental setup is shown in Fig. 1(a). The experiments were conducted in a stainless-steel vacuum chamber of 1200 mm in length and 1000 mm in diameter. The vacuum chamber supplies seven ports, two of which are connected to a turbo-molecular pump and to the coaxial plasma gun, respectively. The gun consists of a pair of coaxial electrodes, which are insulated by a ceramic ring. The outer electrode is made of stainless steel and features a 388 mm long cylinder with an 83 mm inner diameter. The inner electrode is composed of a brass rod and a tungsten–copper hemisphere tip, equipping with a total length of 363 mm and a diameter of 30 mm. The gun is powered by a 150 μF parallel capacitor whose maximum charging voltage is 20 kV.

Plasma was generated by injecting neutral helium gas into the inter-electrode gap through two fast puff valves mounted on the outer electrode. Once the valves open, the gas expanded into the gap at $\sim$1.5 km/s and was then ionized by the high voltage applied across the coaxial electrodes from the capacitor. The axial Lorentz force was produced through the interaction between the radial current carried by the plasma and the azimuthal magnetic field generated by the inner electrode, which rapidly accelerated the plasma. After the current-carrying plasma reached the tip of the inner electrode, it inclined toward the centerline, developing an axial component of current, thus forming a Z-pinch. Between successive discharges, the ambient pressure was maintained at 10^{-4} Pa to minimize the influence of background pressure and improve shot-by-shot repeatability.

Five operating conditions, denoted as (i)–(v) and illustrated in Table I, were used for plasma generation to study the influence of operating parameters on pinch formation. The time interval between the valves start and the capacitor discharge initiation, namely, the gas injection time, was selected between 160 and 340 μs. The lower limit was the shortest injection time allowed for the gas breakdown, while the upper limit corresponded to the maximum accelerated mass. The capacitor charging voltage varied from 7.5 to 10.5 kV, as voltages below this range failed to ionize the injected gas, while higher voltages compromised the discharge repeatability.

B. Diagnostics

The diagnostics and their diagnostic areas are presented in Figs. 1(a)–1(c). As shown in Figs. 1(b) and 1(c), a cylindrical coordinate system (r, θ, z) was established, the z axis coincided with the electrode axis, and the origin was at the center of the muzzle plane. Optical emission from the plasma jet was detected by two photodiodes (Thorlabs PDA-10A), each coupled to an optical fiber with a 200 μm core diameter and a collimator. Each collimator consisted of a convex lens (25.4 mm in diameter and 75.0 mm in focus length) and an aperture limiting the effective lens diameter to 12 mm. This configuration resulted in a 0.08° angle of view. The corresponding optical collection areas on the z axis had a diameter of 14 mm and were centered at $r = 0$, $z = 120$ mm and $r = 0$, $z = 180$ mm, respectively.

Azimuthal magnetic field measurements were obtained using an axial array of magnetic probes, which consists of six chip-inductor probes spacing 10.5 mm each. The array allowed simultaneous sampling of the azimuthal magnetic field at six axial positions. By radially moving the probe array, two-dimensional profiles of the azimuthal magnetic field were constructed over the region $r = 10$–34 mm and $z = -15$–37.5 mm. All probes were calibrated using a Helmholtz coil, yielding a calibration coefficient of 0.51 V/T with an uncertainty of less than 5%.

The discharge current I_{gun} was measured at the base of the gun with a current transformer (Pearson 101). These optical, magnetic, and current signals were recorded by an eight-channel digital oscilloscope (PicoScope 4824A).

The momentum distribution of the ejected plasma was characterized using a radial array of nine ballistic pendulums, and each pendulum was suspended on a sleeve bearing by a thin tungsten rod with an end surface perpendicular to the z axis, whose structure is illustrated in Fig. 2(a). The plasma momentum p imparted to each pendulum was inferred from its maximum deflection angle α, recording by a digital camera. Based on conservation of angular momentum and energy, the relations,

$$\frac{I\omega^2}{2} = (M_p + M_r) g l_{cm} (1 - \cos\alpha), \tag{1}$$

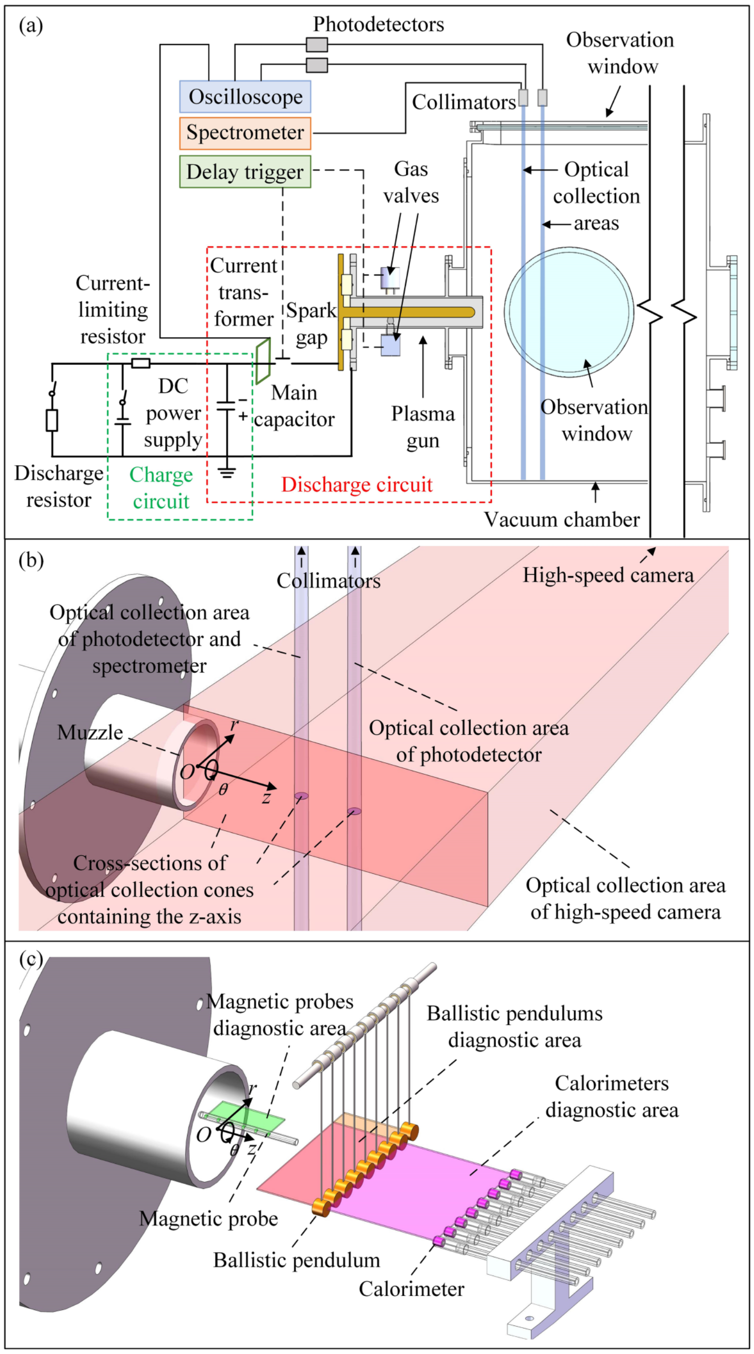

FIG. 1. (a) Schematic of the experiment setup, showing the circuit, the coaxial plasma gun, the vacuum chamber, and part of the diagnostics. (b) Optical collection areas of the photodetectors, spectrometer, and high-speed camera. The collimators and the high-speed camera placed outside the chamber are not shown for clarity. (c) Magnetic probes, ballistic pendulums, calorimeters, and their diagnostic area. Each diagnostic device was used separately in each shot.

$$I\omega = pl_p = \eta_m S_p l_p, \tag{2}$$

were used, where I is the inertia momentum of the pendulum-rod system, ω is the maximum angular velocity of this system, l_{cm} is the distance from the center of mass of this system to the pivot, l_p is the distance from the center of the pendulum to the pivot, M_p is the pendulum mass, and M_r is the wire mass. The momentum density η_m was obtained by dividing the plasma momentum by the pendulum face area S_p. Measurements were performed from $r = -64$ mm to $r = 64$ mm at $z = 70$ and 120 mm.

Energy deposition was evaluated using a radial array of eight calorimeters. As shown in Fig. 2(b), each calorimeter consisted of a

TABLE I. Basic parameters of the five operating conditions used in this paper.

Operating condition	Gas injection time (μs)	Capacitor voltage (kV)	Peak current (kA)
(i)	200	7.5	126
(ii)	200	9	151
(iii)	200	10.5	177
(iv)	160	9	147
(v)	340	9	155

tungsten thermal collector with one end facing the plasma flow, and the opposite end bonded to a K-type thermocouple. The absorbed energy density q_{abs} was determined from the temperature rise ΔT, which value was recorded by an eight-channel temperature logger (Pico TC-08). According to the equation,

$$q_{abs} = \frac{4c_W M_c \Delta T}{\pi d^2}, \tag{3}$$

where c_W is the specific heat of tungsten and M_c and d are the mass and diameter of the collectors, respectively. The profile of the absorbed energy density from $r = -64$ mm to $r = 48$ mm and $z = 70$ mm to $z = 220$ mm was obtained by axial scan of this array. Radiative contributions to the measured energy absorption, including thermal radiation from the collectors to the surroundings and optical radiation from the plasma incident on the collectors, were evaluated and found to be negligible in this experiment. The measured energy absorption was attributed to the kinetic energy of the particles in the plasma flow.

Spectroscopic measurements were carried out using a spectrometer (ANDOR SR-750-A) coupled with a fiber and collimator. The optical collection area was located at $r = 0$, $z = 120$ mm, coinciding with that of the photodiode measurement, and the corresponding exposure was 18–30 μs, covering the pinch evolution. The electron density n_e was calculated from the Stark width of the H_β line, according to[23]

$$n_e = \left(5 \times 10^{10} \times w_s\right)^{1.5}, \tag{4}$$

where n_e is in m^{-3} and w_s is the Stark width in nm. Under the conditions of this study, the Doppler broadening of the H_β line was estimated to be negligible for determining w_s. Therefore, the Gaussian component of the Voigt profile was taken as the instrument broadening, which was measured to be 0.20 nm using a Hg lamp. The

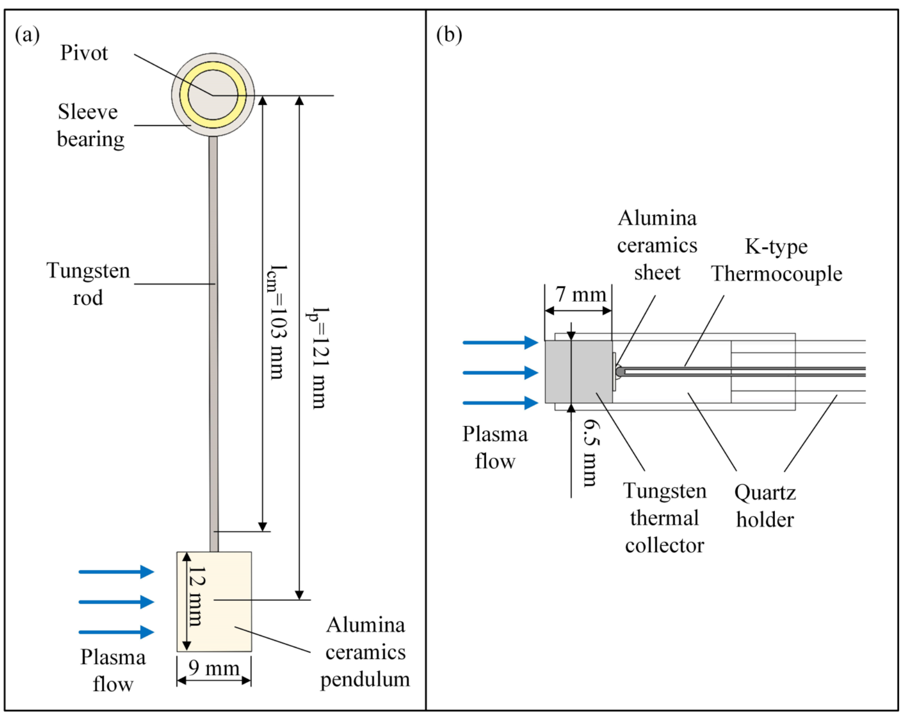

FIG. 2. Cross sections of (a) a ballistic pendulum and (b) a calorimeter. The blue arrows denote the direction of the plasma flow generated by the gun.

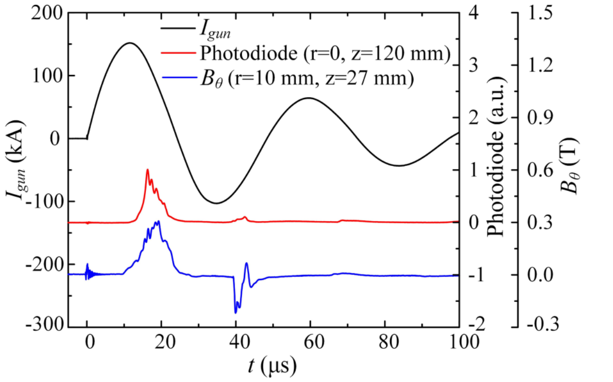

FIG. 3. Typical waveforms of the discharge current, optical emission, and azimuthal magnetic field recorded under operating condition (ii). Each pulse of optical and magnetic signals corresponds to a pulse of plasma jet. The similarity between the optical and magnetic signals implies that the plasma luminance is related to the azimuthal magnetic field.

Stark width w_s, corresponding to the Lorentzian component, was then obtained from Voigt fitting with the Gaussian width fixed.

Time-resolved plasma evolution was visualized using a high-speed camera (Phantom VEO410L) through a side window of the chamber. Images were acquired at 340 000 frame/s with a spatial resolution of 128×48 pixels and an exposure time of 0.93 μs.

III. RESULTS AND DISCUSSION

A. Typical characteristics of the plasma

Representative current, optical, and magnetic signals obtained under operating condition (ii) are presented in Fig. 3. The time t was defined as zero at the initiation of discharge. The discharge current I_{gun} exhibited a damped sinusoidal waveform, which had a half-period of approximately 24 μs and a total duration of over 100 μs. Optical emission measurements revealed three distinct pulses, corresponding to three ejections of plasma during the first three half-periods of the I_{gun} oscillation. The appearance of multiple plasma pulses within a single shot is commonly observed in coaxial plasma gun experiments. It is generally attributed to the reconstruction of current paths within the gun after I_{gun} reversal and the succeeding axial acceleration of the plasma.[24] As the discharge progressed, both the current amplitude and the available neutral gas decreased, causing the emission intensity of the plasma to decrease and become nearly undetectable after the first three pulses in this experiment. The first optical pulse was much larger than the other two in magnitude, indicating that the majority of the plasma energy was carried by the first ejection. The waveform of the azimuthal magnetic field B_θ varied nearly synchronously with the optical signal, especially for the first peak, which not only displayed a similar structure, and possessed a major part as well. The profile of B_θ, averaged over three to five repeated shots with a standard deviation within 15%, is shown in Fig. 4. B_θ was concentrated in the regions near the outer electrode edge and close to the z axis, suggesting that these regions were associated with the pinch effect.

High-speed images of the plasma jet under five operating conditions are shown in Fig. 5. For the first four conditions, a bright plasma column was observed downstream of the muzzle, which had a minimum diameter of below 10 mm and a length of over 100 mm. The column formed and prolonged following the plasma ejection and then diminished as I_{gun} decreased. These observations indicated that the plasma continuously ejected from the inter-electrode gap was coupled with the discharge circuit and underwent radial pinching, which is a typical feature of the flow Z-pinch. Increasing the capacitor charging voltage enhanced the column brightness, consistent with the higher density in excited particles and the higher intensity in the pinch effect. Reducing the gas injection time led to a decrease in column diameter, suggesting stronger radial compression. Under the final condition, an overlong gas injection time resulted in a relatively uniform ejection across the muzzle plane, and the plasma column became much dimmer in comparison with the first four conditions, even though the aperture of the camera was increased.

B. Profile of current

The axial current I_z was calculated from the measured B_θ using Ampere's circuital law. Since the axial symmetry of the plasma jet was shown in the high-speed images, and the measured

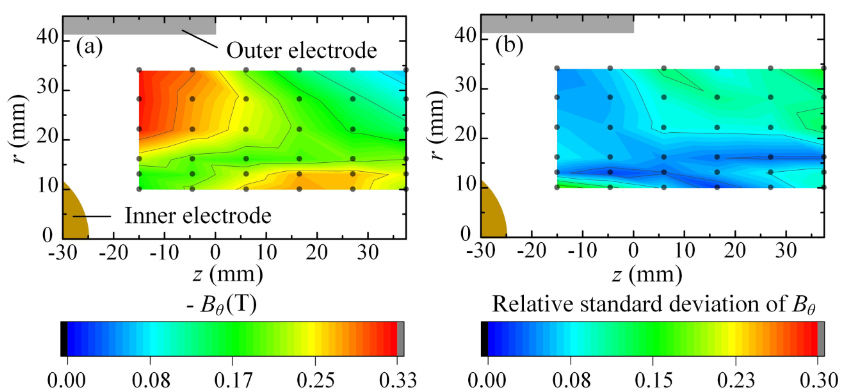

FIG. 4. (a) The profile of azimuthal magnetic field and (b) its shot-to-shot standard deviation at $t = 18\,\mu$s when the azimuthal magnetic field reached its peak under operating condition (ii). The black dots denote the magnetic measurement points used to construct the two-dimensional profiles. The azimuthal magnetic field is shown to concentrate toward the z axis downstream of the muzzle.

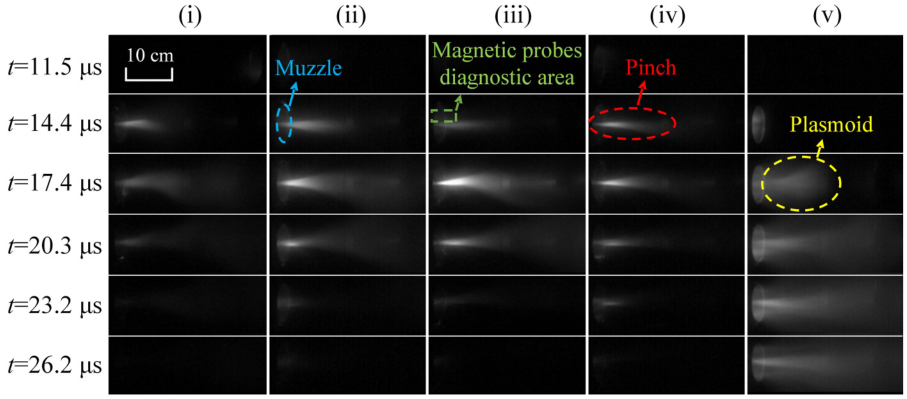

FIG. 5. Side-on view of the plasma evolution under the five conditions. The camera aperture setting was 3 for conditions (i)–(iv) and 5 for condition (v). The blue circle denotes the edge of the outer electrode, and the green rectangle indicates the area with the same r and z coordinates as the magnetic probe diagnostic area. For conditions (i)–(iv), the bright columnar structure marked by the red circle is identified as the Z-pinch. In condition (v), the plasma forms a plasmoid-like structure marked by the yellow circle.

difference between $B_\theta(r_m, z_n)$ and $B_\theta(-r_m, z_n)$ was within 30% in magnitude, the B_θ was treated as axisymmetric, giving the relation,

$$I_z = \frac{2\pi r B_\theta}{\mu_0}, \tag{5}$$

where $I_z(r_m, z_n)$ is the axial current enclosed within radius r_m at axial position z_n.

The resulting current profiles are shown in Fig. 6. In all five conditions, the highest I_z was observed near the outer electrode edge. For the first four cases, I_z conducted to the edge accounted for approximately 40% of the peak I_{gun}, whereas under the longest injection condition, the fraction decreased to only 18%. This reduction was attributed to the excessive injection of gas mass, which postponed the plasma ejecting time and thereby lowered I_{gun} during the ejection. In addition, the accumulation of plasma in the inter-electrode gap created a low-impedance current path upstream of the muzzle, further reducing I_z observed at the exit. A radial concentration of I_z downstream of the muzzle was also observed in the first four conditions (red dashed boxes), which will be more obvious in the current density profiles.

The current density components were determined from the measured B_θ profiles as follows:

$$J_r = -\frac{1}{\mu_0 r}\frac{\partial (r B_\theta)}{\partial z}, \tag{6}$$

$$J_z = \frac{1}{\mu_0 r}\frac{\partial (r B_\theta)}{\partial r}. \tag{7}$$

The azimuthal component was neglected due to the absence of the kink instability within the measurement region.

As shown in Fig. 7, the current flowed from the outer electrode to the inner electrode, forming closed loops that extend over 37.5 mm downstream of the muzzle. A pronounced concentration of current density near the z axis was observed, though the profile varied with operating condition. To quantify the degree of current concentration,

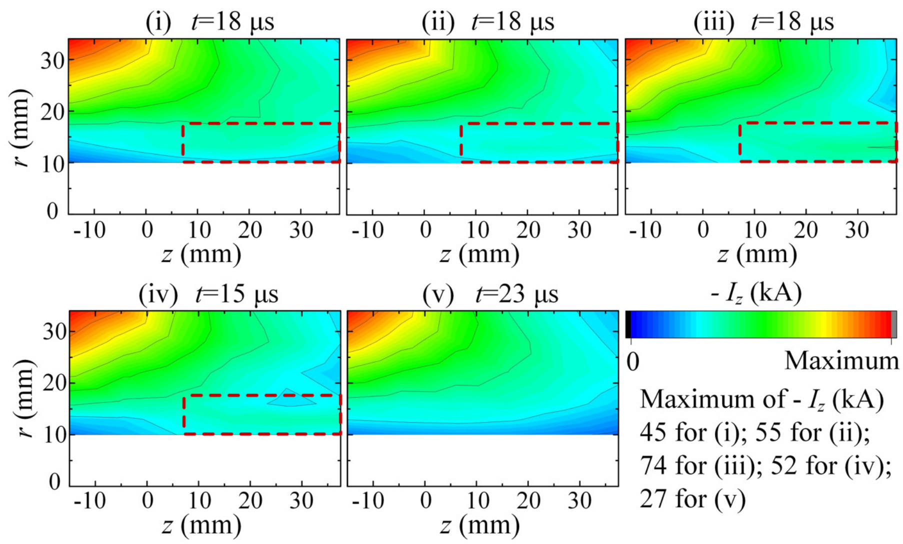

FIG. 6. Axial current profiles under the five conditions. For each condition, the time corresponds to the moment when the azimuthal magnetic field reached its peak, and the same applies to Figs. 7–10. Under the first four conditions, the axial current exhibits a tendency of radial concentration in the area highlighted by the red dashed rectangles.

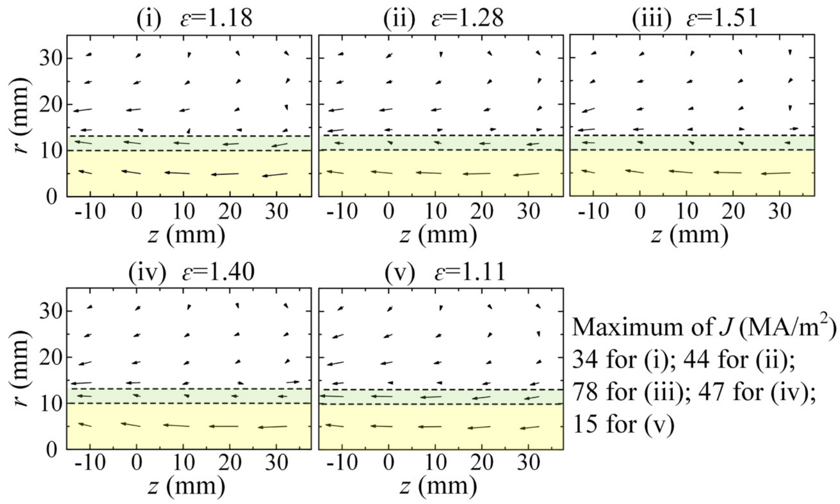

FIG. 7. Profiles of current density. The yellow and green rectangles indicate the regions of $r \leq 10$ mm and $r \leq 13$ mm, where axial current density is concentrated. The parameter ε, quantifying the radial compression of the axial current density, is defined as $\varepsilon = I_z(r_1, z_1)r_2^2/I_z(r_2, z_1)r_1^2$, where r_1 = 10 mm, r_2 = 13 mm, and z_1 = 27 mm. The radial concentration of current density is apparent, especially in the first four conditions.

a parameter ε was defined as the ratio of average axial current density within $r \leq 10$ mm to that within $r \leq 13$ mm at $z = 27$ mm. Increasing the capacitor voltage led to an enhancement of both ε and J_z, while an overlong gas injection time decreased both of them. For the condition (iv) with the shortest injection time, the maximum J_z was close to that measured in condition (ii), but a much higher ε was obtained.

The CDV equilibrium Z-pinch model,[16] assuming the current density to be proportional to the electron density, was adopted to determine the relative electron density. Since J_z dominated the current density near the z axis, ε was reasonably regarded as the ratio of average electron density within the two regions, positively correlated with pinch intensity. The observed variation of ε across operating conditions, thus, indicated that higher capacitor voltage and shorter gas injection time enhanced radial plasma compression.

The concentration of the current and particles provided clear evidence of pinch formation. The radial compression resulted in a higher collision frequency of particles, increasing the density of excited particles, and thereby forming a bright region as shown in Fig. 5. A variation of the pinched current density along the z axis was also detected. The maximum of J_z was measured at $z = 16.5$ mm to $z = 27$ mm, rather than immediately downstream of the inner electrode, which corresponded well to the bright region observed in Fig. 5. This was because of the delay of the plasma response to the radial Ampere force and will be discussed in Sec. III C.

C. Plasma dynamics

The Lorentz force density $\boldsymbol{F}$ was calculated from the measured B_θ. From magnetohydrodynamic (MHD) theory,[25] the relation,

$$\boldsymbol{F} = \boldsymbol{j} \times \boldsymbol{B} = \frac{1}{\mu_0}(\nabla \times \boldsymbol{B}) \times \boldsymbol{B} = \frac{1}{2\mu_0}\frac{\partial B^2}{\partial s}\boldsymbol{t} + \frac{B^2}{\mu_0 R_c}\boldsymbol{n} - \nabla\left(\frac{B^2}{2\mu_0}\right), \tag{8}$$

applied, where $\boldsymbol{t}$ is the tangent, $\boldsymbol{n}$ the normal, s the curvilinear coordinate, and R_c the curvature radius of the magnetic field line. Since the B_r and B_z were proven to be negligible compared with B_θ for the Z-pinch plasma generated by a similar device,[26] $\boldsymbol{B}(r, \theta, z)$ was regarded as $B_\theta(r, z)\hat{\boldsymbol{\theta}}$ based on the approximation of axisymmetric B_θ. A simplified expression of $\boldsymbol{F}$ was then obtained as follows:

$$\boldsymbol{F} = -\frac{B_\theta^2}{\mu_0 r}\hat{\boldsymbol{r}} - \frac{1}{2\mu_0}\frac{\partial B_\theta^2}{\partial r}\hat{\boldsymbol{r}} - \frac{1}{2\mu_0}\frac{\partial B_\theta^2}{\partial z}\hat{\boldsymbol{z}}. \tag{9}$$

The profile of radial and axial force components, F_r and F_z, are presented in Figs. 8 and 10, respectively.

As shown in Fig. 8, negative F_r components dominated the measurement region, while positive ones were comparatively minor in both magnitude and extent. Two regions of strong negative F_r, inter-electrode area adjacent to the muzzle [region (I), orange boxes] and the axial area downstream of the inner electrode [region (II), green boxes], both contributed significantly to pinch formation.

An estimation was made to quantify the influence of $\boldsymbol{F}$ on the motion of the plasma. The mass density of the plasma within the inter-electrode gap was calculated by the following relations:

$$p_c = \frac{\mu_0}{4\pi}\ln\frac{r_{\text{out}}}{r_{\text{in}}}\int_0^{\frac{T}{2}} I_{\text{gun}}^2\, dt, \tag{10}$$

$$m_c = \frac{p_c}{v_z}, \tag{11}$$

$$V_c = \pi\left(r_{\text{out}}^2 - r_{\text{in}}^2\right)v_z\frac{T}{2}, \tag{12}$$

$$\rho_c = \frac{m_c}{V_c}, \tag{13}$$

where p_c, m_c, V_c, v_z, and ρ_c denote momentum, mass, volume, axial velocity, and mass density of the plasma, respectively. r_{out} is the inner radius of the outer electrode, and r_{in} is the radius of the inner electrode. The upper limit of the integral is chosen as $T/2$, the first half-period of I_{gun}, to yield an averaged density during the first ejection. Typically, under condition (ii), v_z was measured as 70 km/s,

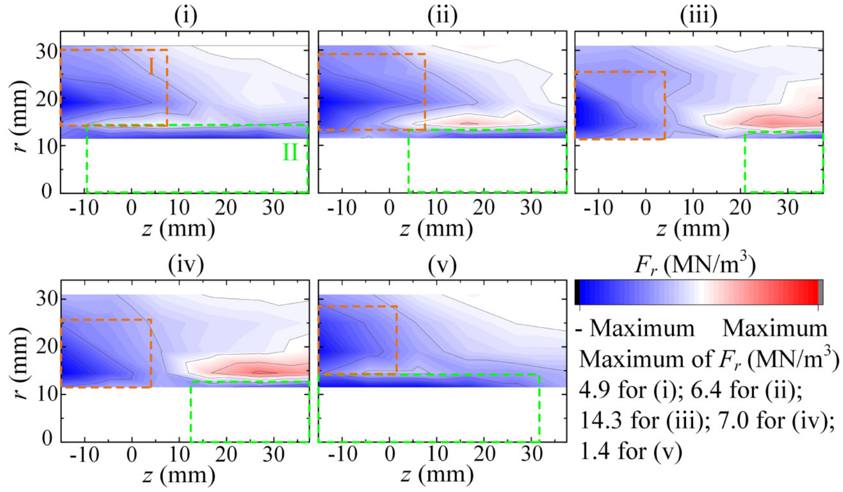

FIG. 8. Profiles of radial Lorentz force density. Profiles in each condition exhibit two distinct regions where strong negative radial force is present. These two types of regions are marked by the orange boxes and green boxes and are named as region (I) and region (II), respectively.

the resulting ρ_c was 5×10^{-5} kg/m^3, corresponding to a radial acceleration of -1×10^{11} m/s^2 under a -5 MN/m^3 F_r. Assuming the plasma entered region (I) with a velocity $\boldsymbol{v} = v_z\hat{z}$, a 40 mm axial motion produced a 16.3 mm radial displacement, allowing the plasma to concentrate on region (II) from the inter-electrode gap. While an insufficient axial motion was not able to make a pronounced radial displacement, thus formed a hollow region approximately 40 mm in length downstream of the inner electrode.

Once transported into region (II), the plasma underwent a stronger F_r, leading to Z-pinch formation. Z-pinch plasmas generated in similar devices have been shown to approach radially equilibrated,[16] and the pressure profile $P(r)$ was determined by

$$\frac{\mathrm{d}P(r)}{\mathrm{d}r} = F_r(r) = -\frac{B_\theta}{\mu_0 r}\frac{d(rB_\theta)}{dr}. \tag{14}$$

Therefore, the determination of the pressure profile requires the magnetic field profile in the core of the Z-pinch plasma. However, it was hard to acquire repeatable magnetic probe signals within $r < 10$ mm, direct construction of B_θ in the core was not feasible. To estimate the plasma parameters in this region, the Bennett pinch profile,[25] which satisfied Eq. (14) and has been used in a similar device,[27] was introduced as a model approximation. The Bennett profile of azimuthal magnetic field B_θ, current density J, and pressure P were calculated as follows:

$$B_\theta(r) = \frac{\mu_0 I}{2\pi}\frac{r}{r^2 + a^2}, \tag{15}$$

$$J(r) = \frac{I}{\pi}\frac{a^2}{(r^2 + a^2)^2}, \tag{16}$$

$$P(r) = \frac{\mu_0 I^2}{8\pi^2}\frac{a^2}{(r^2 + a^2)^2}. \tag{17}$$

The parameters I and a were determined by B_θ measured at $r = 10$ mm, $z = 27$ mm and $r = 13$ mm, $z = 27$ mm, yielding radial profiles of B_θ, J, and P as shown in Fig. 9. Raising the capacitor voltage increased the current carried by the pinched plasma, enhanced F_r, while reducing the gas injection time decreased the pinched mass,

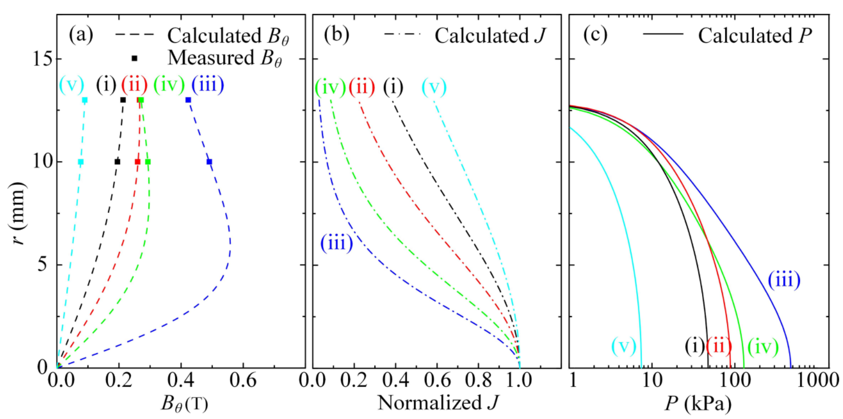

FIG. 9. Radial profiles of (a) azimuthal magnetic field, (b) current density, and (c) pressure. The solid squares in subfigure (a) are the azimuthal magnetic fields measured by magnetic probes. The curves were obtained from the Bennett profile, whose parameters were calculated from the measured magnetic field. The colors black, red, dark blue, green, and light blue correspond to conditions (i)–(v), respectively. These profiles show that the conditions with stronger radial Lorentz forces produce more concentrated distributions in current density and pressure.

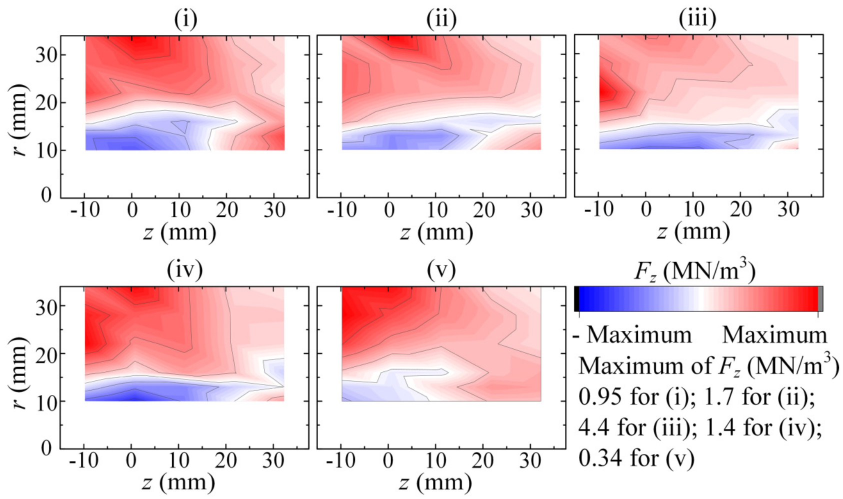

FIG. 10. Profiles of axial Lorentz force density. Profiles in each condition show apparent radial shears of axial Lorentz force density, indicating significant variation of the axial acceleration at different r coordinates.

making the plasma more compressible. Both effects led to a more central concentrated J and, thus, resulted in a higher P on the axis.

The above analysis demonstrated that the Z-pinch formation included two stages: a radial concentration in the inter-electrode area, followed by an intensive magnetic compression on the axis. Moreover, the pinch intensity could be modulated by adjusting operation parameters.

Figure 10 illustrates the profile of shearing F_z, which was negative near the axis and became positive at larger radial positions. The negative F_z was generated from the interaction between the negative B_θ and the positive J_z, which was oriented from the current continuity in the hollow region downstream of the inner electrode. Typically, a difference in F_z between $r = 34$ mm and $r = 10$ mm was ~ 1 MN/m^3. Taking $\rho_c = 5 \times 10^{-5}$ kg/m^3, a 0.5 μs action of the shearing F_z resulted in a 10 km/s difference of axial velocity, which was comparable to the bulk axial velocity. The observed stability in the measurement region during the Z-pinch lifetime (as shown in Fig. 5) was, therefore, consistent with the presence of stabilizing shear flow.

D. Plasma parameters

Figure 11(a) shows the electron density n_e obtained from the Stark width of the H_β line, together with the bulk axial velocity v_z determined from the time-of-flight method of two axially separated optical signals. The measured electron densities and bulk velocities were at the order of 10^{22} m^{-3} and 10–100 km/s, respectively. As the capacitor voltage increased from 7.5 to 10.5 kV, the energy stored increased by 96% and the peak discharge current rose by 40%; however, the electron density on the axis increased by 3.3 times. This dramatic rise correlated with both the higher degree of ionization and the more pronounced radial compression of the plasma. Similarly, as the gas injection time decreased from 340 to 160 μs, the electron density rose by 3.0 times, indicating a significant enhancement of the pinch effect. The bulk velocity likewise increased as the voltage rose and the injection time decreased, consistent with a stronger axial Lorentz force within the gun and a reduced plasma mass.

Using the measured n_e and v_z, the momentum density η and kinetic energy density q of the plasma were estimated as follows:

$$\eta = \frac{n_e}{Z} m_{\mathrm{He}} v_z^2 t, \tag{18}$$

$$q = \frac{1}{2}\frac{n_e}{Z} m_{\mathrm{He}} v_z^3 t, \tag{19}$$

where m_{He} is the mass of a Helium atom, t is the ejection duration defined by the full width at half maximum (FWHM) of the optical

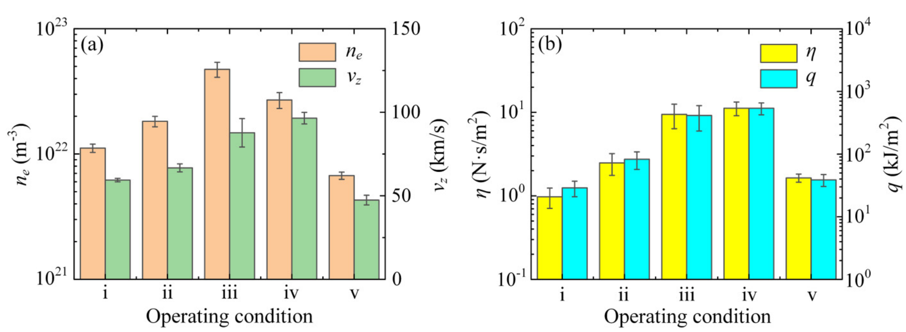

FIG. 11. (a) On-axis electron density at $z = 120$ mm measured by the spectrometer and average bulk axial velocity over $z = 120$–180 mm measured by two photodetectors. (b) Momentum density and energy density derived from the measured electron density and axial velocity.

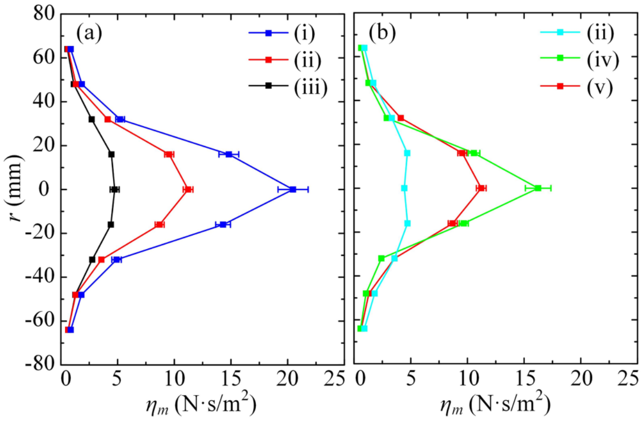

FIG. 12. Radial profiles of momentum density measured by ballistic pendulums at $z = 120$ mm vs (a) capacitor voltage and (b) gas injection time. Conditions (i)–(iv) show central-peak distributions of momentum density, indicating radial concentrations of particles.

emission signal, and Z, the ion charge number, is taken as 1 as a reference value, according to the plasma energy estimation in similar devices.[28] The resulting η and q, shown in Fig. 11(b), are on the order of 1–10 N s/m^2 and 10–1000 kJ/m^2, respectively. Owing to the uncertainty of the assumed Z, these values should be regarded as order-of-magnitude estimations.

Figure 12 shows the momentum density η_m measured by the ballistic pendulums. Although these two methods agreed in order of magnitude, on-axis η_m was higher than η in all five conditions. This discrepancy was possibly due to an overestimation of Z, leading to an underestimation of η. For conditions (iii) and (iv), η_m was closer to η than in the other conditions, suggesting the stronger pinch effect increased Z.

The radial profile of η_m provided direct evidence of radial concentration of the plasma flow, confirming the formation of the flow Z-pinch. The central-peak distribution became more obvious under higher capacitor voltage and lower gas injection time, which corresponded to the variation of pinch intensity with operating conditions. In contrast, condition (v) produced a slightly central-hollow profile, indicative of weak pinching and the sheared axial flow generated by F_z.

Figure 13 shows the profile of the absorbed energy density q_{abs} measured by calorimeters. q_{abs} at the spectrum measurement point was close to the estimated q in conditions (i), (ii), and (v), while it was significantly lower than that in conditions (iii) and (iv). The discrepancy mainly came from the shielding effect of ablated vapor,[28] recycled gas,[29] and bow shock[11] forming upstream of the thermal collectors, which made the energy deposited in the collectors lower than the incident energy. Similar to η_m, the radial profile of q_{abs} showed a central-peak shape in conditions (i)–(iv) and a central-hollow shape in condition (v). As the plasma traveled forward, radial diffusion broadened the beam spot area, reducing the local energy density while increasing spatial uniformity.

E. Applications

Coaxial plasma guns are well-suited for studies of plasma–material interaction owing to their ability to generate plasma jets of high-energy density and short pulse duration. In this experiment, a single coaxial gun produces plasma flows with a wide range of adjustable parameters, enabling multiple processing regimes within the same device.

As the plasma impacts the material, both energy and particles are deposited on the surface, causing rapid changes in surface temperature and composition that can substantially modify the near-surface properties. The temperature evolution within 304 stainless steel after the end of the plasma heat flow was calculated by solving the one-dimensional heat equation and is shown in Fig. 14(a). A typical energy density of 50 kJ/m^2 and a duration of 5 μs, corresponding to conditions (i) and (ii), were used. A high temperature rise exceeding 1500 K persists for over 1 μs, promoting deep ion implantation.[7] A cooling rate of 10^7–10^9 K/s allows recrystallization

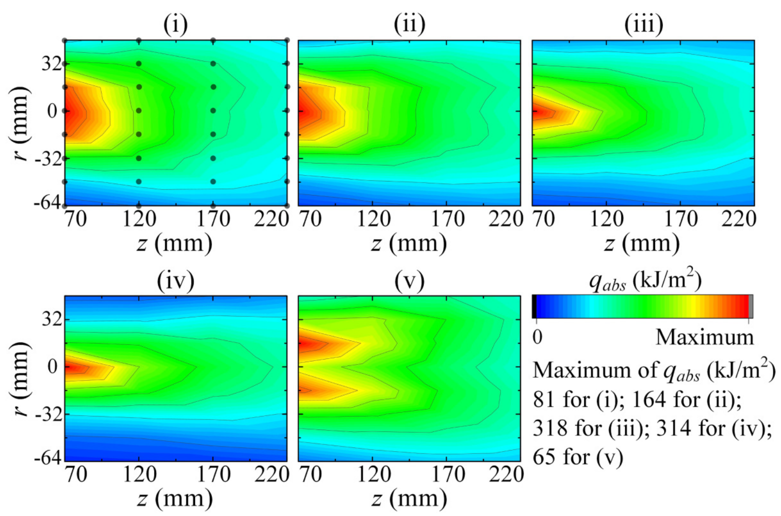

FIG. 13. Profiles of energy density absorbed by calorimeters. Black dots in the profile of condition (i) indicate the calorimeter measurement points used to construct the profile. Conditions (i)–(iv) exhibit central concentrations of energy density, consistent with strong radial compression of the plasma.

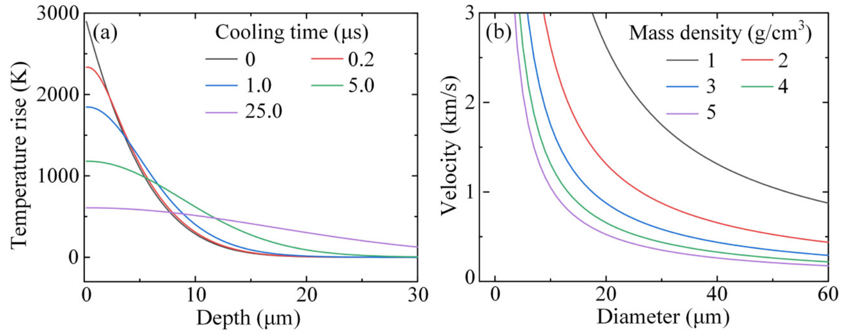

FIG. 14. (a) Temperature evolution within 304 stainless steel, calculated by solving the one-dimensional heat equation. The length of the computational domain is 1 mm. A constant heat flux of 10 GW/m^2 is applied to one boundary for 5 μs and then set to zero. The opposite boundary remains adiabatic. The zero point of the cooling time is defined as the end of the heat flux pulse, and the depth is the distance from the heated boundary. In this figure, a high cooling rate of over 10^7 K/s is achieved, which is essential for coating modification. (b) Calculated velocities of spherical powder particles exposed to a plasma pulse with a momentum density of 35 N s/m^2. The particle velocities are obtained based on the complete transfer of momentum from the interacting plasma to the powder particles. Particles with diameters of tens of micrometers can be accelerated to over 1 km/s, which is beneficial for plasma spraying.

to occur, mixing the elements of the coating and the substrate and thereby enhancing the coating adhesion.[30]

The high axial momentum of the plasma jet also enables efficient acceleration of powder particles for plasma spray. For a typical momentum density of 35 N s/m^2, which was measured under conditions (iii) at z = 70 mm, the velocities of powder particles with different diameters and mass densities were estimated and shown in Fig. 14(b). Velocities exceeding 1 km/s are readily achieved, surpassing those attainable by conventional spraying methods, providing a high coating adhesion without heating the substrate.[31]

In addition, the operation condition (v) provides a large-area and radial uniform plasma flow, which is competent for applications like thin-film deposition,[32] plasma etching,[33] requiring a relatively low energy density of 1–10 kJ/m^2.

IV. CONCLUSION

The formation mechanisms and characteristic parameters of Z-pinch plasma generated by a coaxial plasma gun were investigated through magnetic, optical, and calorimetric diagnostics. Profiles of Lorentz force density, calculated from the measured magnetic field, revealed two distinct regions contributing to pinch formation. Region (I), located near the muzzle within the inter-electrode gap, transported plasma downstream toward region (II), a near-axis zone downstream of the inner electrode. Upon entering region (II), the plasma was intensively compressed by a radial Lorentz force arising from the high axial current density, forming a Z-pinch plasma with high particle density. Measurements from ballistic pendulums and calorimeters confirmed the radial concentration of plasma momentum and kinetic energy resulting from this pinching process. Increasing the capacitor charging voltage enhanced the radial Lorentz force, and reducing the gas injection time decreased the ejected plasma mass. Both changes lead to a more intensive pinching effect and higher bulk velocity, corresponding to a greater and more concentrated momentum and kinetic energy. In contrast, excessive gas injection decreased the radial Lorentz force, thus weakening the pinching effect, leading to a relatively uniform radial profile of momentum and kinetic energy. The peak momentum density of 35 N s/m^2 and the absorbed energy density of 10–300 kJ/m^2 achieved in the present study demonstrated that the device is capable of supporting a range of plasma–material interaction applications, including impulse plasma deposition and pulsed plasma spraying, which will be explored in future experiments. The findings not only advance understanding of Z-pinch formation in coaxial plasma guns but also offer practical guidance for tailoring plasma parameters through control of operating conditions.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jinshuo Zhang: Formal analysis (equal); Investigation (equal); Writing – original draft (equal). **Daren Tang:** Methodology (equal); Validation (equal). **Fantao Zhao:** Investigation (equal); Validation (equal). **Zhen Wang:** Validation (equal); Visualization (equal). **Huijie Yan:** Resources (equal); Writing – review & editing (equal). **Jian Song:** Funding acquisition (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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